

**Address:**

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Education:

**Full Stack Web & App Development
Bootcamp WBS Coding School (online)**

02/2025 – 06/2025

Focused on React, Node.js, JavaScript, and modern web development.

**Bachelor of Science in Electrotechnics
- University of Technology in Kielce
(Poland)**

10/2009 – 02/2013

Specialization: Industrial Automation

Skills:

- MS Office, Git, Bitbucket, Jira
- JavaScript, Node.js, Express, React
- CSS, TailwindCSS, HTML5
- SQL, MongoDB, PostgreSQL, RestAPI, Postman
- Networking and general IT knowledge
- Python, Figma
- Familiar with Docker
- Tia Portal/TwinCat3 (PLC programming)
- WinCC/TwinCat HMI (Visualization)

Soft skills:

- **Problem-Solving:** Analytical mindset with a proven ability to troubleshoot complex automation and software issues under tight deadlines
- **Team Collaboration:** Experienced in working with cross-functional teams to align technical solutions with business objectives.
- **Adaptability:** Quickly learned and applied new technologies (e.g., React, Node.js, JS) during career transition to software development
- **Communication:** Effectively communicated technical concepts to operators, engineers, and stakeholders in multilingual environments
- **Time Management:** Successfully managed multiple projects and deadlines in fast-paced manufacturing and development settings

Languages:

English: C1

German: B2

Polish: Native

Adrian Bakalarz

Full Stack Developer

Profile summary:

Controls Engineer transitioning to Full Stack/Software Developer with 6 years of experience in industrial automation, programming SCADA, HMI, and PLC systems. Leveraged Python and PostgreSQL at Tesla for 3 years to enhance manufacturing efficiency. Applying automation principles—PLCs, motors, and sensors as backend, SCADA/HMI as frontend/UI—to build scalable web applications, with growing expertise in TypeScript, React, and Node.js.

Experience:

Controls Engineer, Tesla, Grunheide

10/2021 – 06/2024

- Updated and improved HMIs for complex machinery to be used by operators and maintenance.
- Implemented code modifications to PLCs to introduce new functionalities, adhering to Object Oriented Programming (OOP) principles. Tracked all changes efficiently using Git for version control.
- Supported commissioning in Berlin and Austin, ensuring seamless system integration.
- Collaborated with cross-functional teams to align control algorithms with production goals.
- Developed Python scripts to analyze sensor data and optimize control algorithms, reducing battery transport crane cycle time from 35 to 30 seconds (14.3% improvement).
- Utilized PostgreSQL to store and query historical process data, enabling rapid fault resolution.
- Designed PostgreSQL queries to monitor equipment performance, collaborating with teams to enhance predictive maintenance.
- Built Python tools for control logic simulation, interfacing with PostgreSQL to validate data integrity.

PLC Programmer, Simtech AP, Krakow (Poland)

09/2018 – 08/2021

- Programmed PLCs (e.g., Siemens Tia Portal) to control machinery, optimizing processes such as material handling and production line in automotive manufacturing, analogous to backend logic development in software applications.
- Developed intuitive SCADA and HMI interfaces for real-time process visualization and operator interaction, mirroring frontend/UI design principles in web development.
- Conducted system diagnostics and debugging of PLC code, resolving automation faults under tight deadlines, improving system uptime.
- Created detailed user guides and maintenance protocols for SCADA and HMI systems, enhancing operator training and system maintainability, comparable to writing technical specs for software projects.
- Documented code changes and system configurations, maintaining clear records for future upgrades, aligning with version control practices like Git in software development.